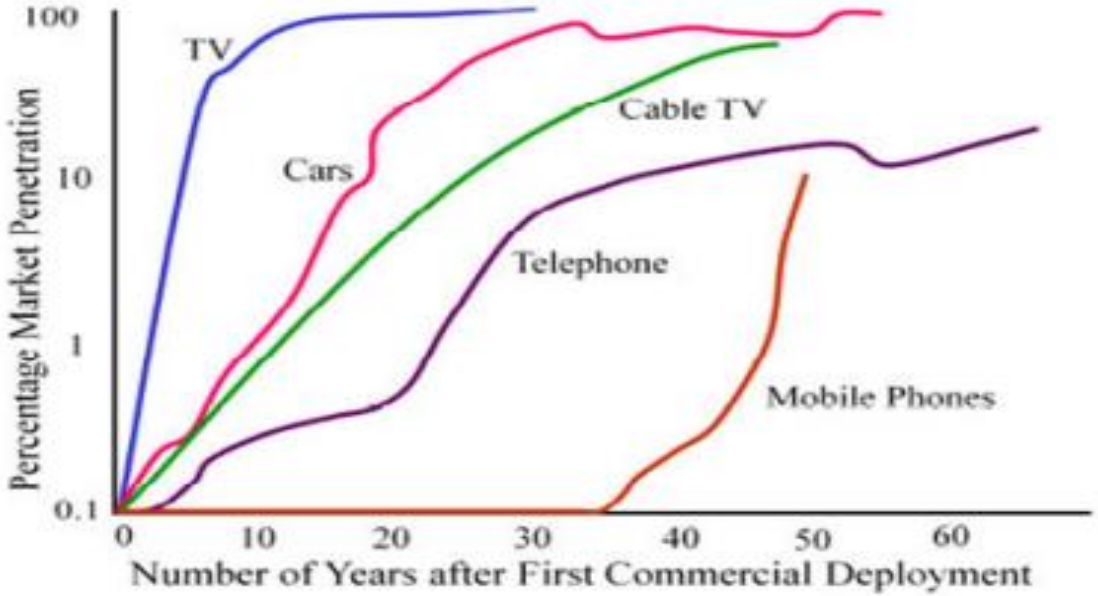
Long Term Evolution (LTE)
TCS 411

Abdulkarim A. Oloyede

Early Developments

- 1897: Marconi provides contact with ships in English Channel.
- 1934: 5000 AM mobile phones in USA (police).
- 1935: Edwin Armstrong invents FM.
- 1946: First public mobile telephone system introduced in 25 cities.
- 1950s: First mobile users connected to the PSTN via IMTS.
- 1970s: Cellular concept born.

- Mobile phones took 50 years to reach 10% market penetration.
- Early phones were large and heavy.
- Limited semiconductor and battery technology.
- Hostile radio channel.



- Motorola DynaTAC (1983) - 800g



- Nokia 3310 (2000) —N1000 , 133g



- Nokia 1011 (1992) – 475g

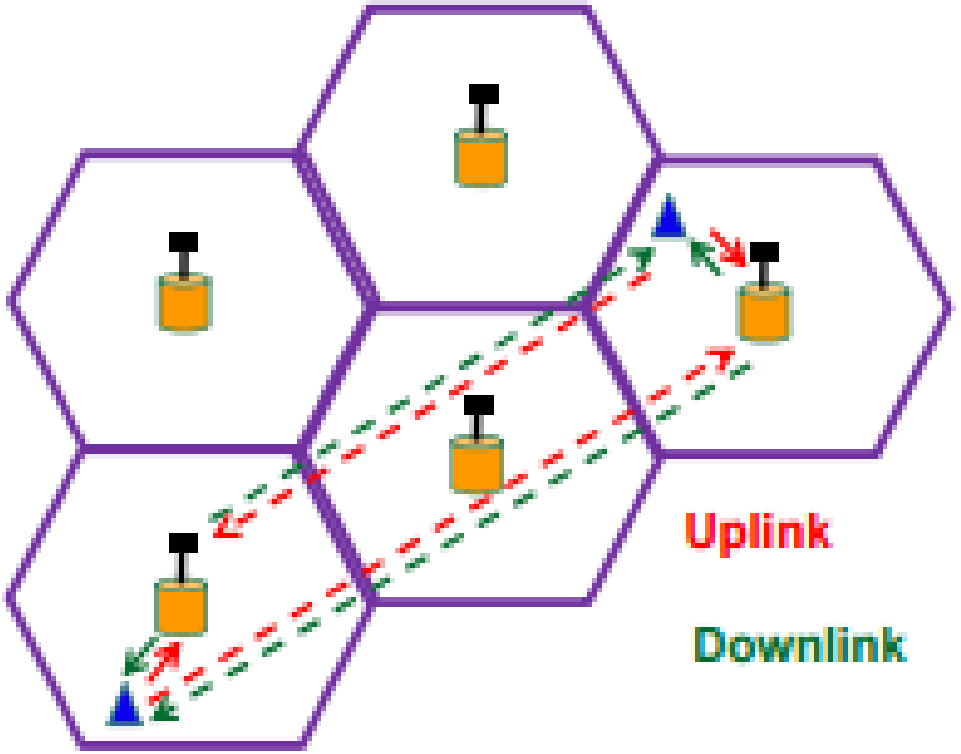


- iPhone (2007) —>N100,000, 135g



Cellular Principle

- Multiple transmitter per city
- Frequency reuse in a planned way across cells



Why Cellular

- To provide sufficient capacity through efficient frequency re-use.
  - Applies to FDMA/TDMA systems.
  - Need to ensure co-channel users are far enough apart.
- Low power transmissions save battery life and ameliorate potential health issues.
- Alternatives
  - Large transmit towers and lots of radio spectrum
  - Ad-hoc arrangements of base stations and frequency assignments.

First Generation - America

- **Advanced Mobile Phone System (AMPS) - 1983**

Developed by Bell Labs

- **Analogue, Frequency Modulation (FM)**

666 channels in a 40 MHz allocation centred around 800 MHz, each 30 KHz wide

Licenced for two network operators

Growing demand led to additional frequencies being assigned

832 channels in total

- **No protection against eavesdropping**

First Generation - Europe

- **Many standards deployed in the early 1980s**

Roaming not possible

Analogue, frequency modulation (common)

- **European Total Access Cellular System (ETACS) in the UK**

1320 channels in a 66 MHz allocation centred around 900 MHz, each 25 KHz wide

Operational from 1983 until 2001

- **No protection against eavesdropping**

Motivation for Second Generation

- Spectral efficiency
- Security
- More reliable speech quality
- New services such as SMS, caller-ID etc...
- International roaming (in Europe)
- Longer battery life
- Network access from mobile or fixed handset

Second Generation - America

- **Digital AMPS (IS-54 and then IS-136)**

FDMA/TDMA systems with 30 KHz channels, 3 users per channel
Compatible with existing systems

IS-136 added features such as text messaging

- **IS-95 (cdmaOne Is the brand name)**

Developed by Qualcomm

This is the first CDMA based 2G digital cellular standard

CDMA preforms well against(Narrow band) Interference and (multipath) signal fading

- Lunched in 1995 in Hong Kong
- Supports voice or data up to a bit rate of 14.4kbps
- **Roaming difficult in the US, limited encryption**

Second Generation - Europe

- **Global System for Mobile Communications (GSM)**

Originally Groupe Specialé Mobile

FDMA/TDMA system with 200 KHz channels, 8 users per channel
Compatible with existing systems

Included novel features such as SMS

- **Pan-European standard**

Developed by the European Telecommunications Standards Institute (ETSI)

Same spectrum assigned across Europe

- **Roaming easy throughout Europe, but expensive!**

Motivation for Third Generation

- **Less clear...**

Second generation was very successful

Most users happy with second generation (and still are)

- **Higher data rates for more demanding applications**

Improved mobile Internet access

Video telephony, video streaming

Mobile TV?

Data applications have certainly taken off, placing a lot of strain on the networks.

3 minute conversation ~= 100 kbytes

3 minute video clip ~= 7 Mbytes

Third Generation Systems

- Universal Mobile telecommunication System (UMTS)
- 3rd Generation 3G mobile cellular system
- Evolution of GSM (2G)
- Supports UTRA (UMTS terrestrial Radio Access) supports several different terrestrial air interface Multi user Access in UTRA can be supported by UTRA-FDD or UTRA TDD (Time Division Duplex)
- **W-CDMA (Wideband Code Division Multiple Access)**

Air interface based on FDD and CDMA (Europe, Japan)

- **TD-CDMA (Time Division Code Division Multiple Access)**

Air interface based on TDD and TDMA/CDMA (Europe)

- **TD-SCDMA (Time Division – Synchronous Code Division Multiple Access)**

Air interface based on TDD and TDMA/CDMA (China)

- **CDMA2000 (multicarrier CDMA)**

Developed by Qualcomm (America)

Competes with UMTS

Issues with 3G

- **What happened to a single global standard?**
- Licence fees

Similar network build costs

GPRS and EDGE provide good data rates at much lower cost (using GSM)

Applications

YouTube, iPlayer, others?

Coverage is still sparse

Long Term Evolution (LTE)

- 3GPP-LTE (3.9G)
- Completely new system based on IP
- 100Mbit/s peak transmission rate
- Doesn’t meet ITU-R specification for 4G (1Gbit/s)
- LTE-Advanced does

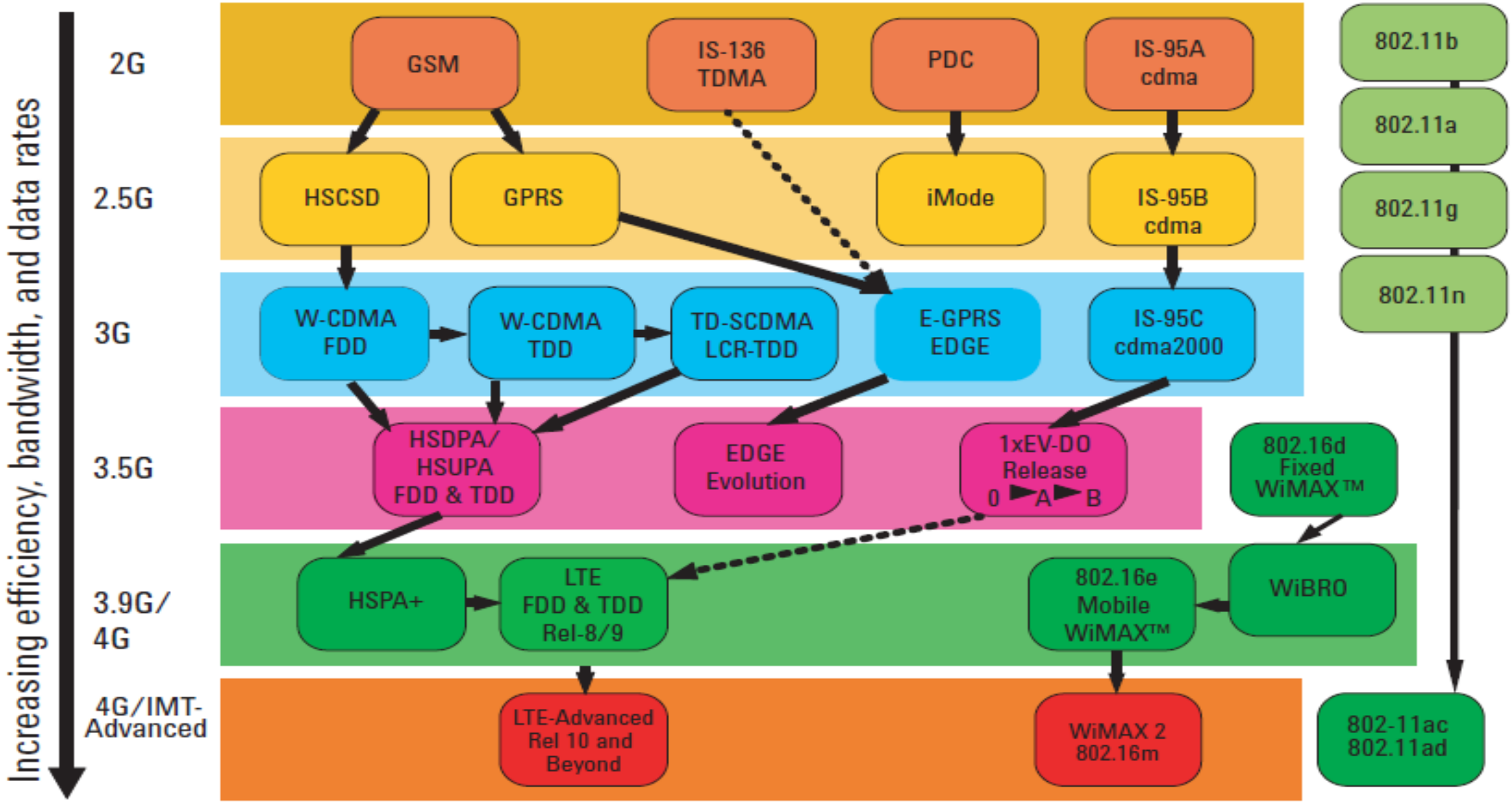


Figure 1. Wireless evolution 1990–2011 and beyond

Table 1. Evolution of UMTS specifications

Release	Functional Freeze	Main Radio Features of the Release
Rel-99	March 2000	UMTS 3.84 Mcps (W-CDMA FDD & TDD)
Rel-4	March 2001	1.28 Mcps TDD (aka TD-SCDMA)
Rel-5	June 2002	HSDPA
Rel-6	March 2005	HSUPA (E-DCH)
Rel-7	Dec 2007	HSPA+ (64QAM DL, MIMO, 16QAM UL), LTE & SAE feasibility study, EDGE Evolution
Rel-8	Dec 2008	LTE work item – OFDMA air interface, SAE work item, new IP core network, 3G femtocells, dual carrier HSDPA
Rel-9	Dec 2009	Multi-standard radio (MSR), dual cell HSUPA LTE-Advanced feasibility study, SON, LTE femtocells
Rel-10	March 2011	LTE-Advanced (4G) work item, CoMP study, four carrier HSDPA

Summary

- **Cellular networks have been around for 30 years, but mass-market for the last 20 years.**

First generation systems were based on analogue transmission and designed for speech

Second generation employ digital speech coding and transmission to support speech and data

Third generation is designed to provide higher data rates for more demanding applications

- **Several standards were produced for first, second and third generation with variable success**
- **3GPP-LTE is successful but has competition**